

(<https://osf.io/ktz5h/>). These cohort studies offer opportunities to assess prospectively whether there are long-term effects of COVID-19 on brain development in children and adolescents and risk for ADHD and other neuropsychiatric disorders. Further, these studies could evaluate the hypothesis that ADHD may be a risk factor for COVID-19, as prevention methods (social isolation and systematic handwashing and mask-wearing) require attention and vigilance, making compliance difficult for children with ADHD. Prospective monitoring could also evaluate the hypothesis that effective treatment with stimulant medication may provide children with ADHD some protection from acquiring COVID-19.

In summary, based on Fauci's suggestion 10 years ago of the importance of lessons learned from the 1918 influenza pandemic and Wender's suggestion 50 years ago that residual sequelae of the 1918 pandemic were manifested as minimal brain dysfunction in some children, it may be worthwhile to consider the hypothesis that the COVID-19 pandemic may result in a novel etiologic subtype of ADHD that clinicians may recognize in patients in the future.

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Supporting Children With Neurodevelopmental Disorders During the COVID-19 Pandemic



To the Editor:



Families of children with neurodevelopmental disorders are especially vulnerable during the COVID-19 pandemic. Physical distancing requirements and closure of schools and services in the context of the COVID-19 pandemic are likely challenging to everyone but may be particularly impactful for families with children with neurodevelopmental disorders ([NDDs], eg, intellectual disability, attention-deficit/hyperactivity disorder [ADHD], autism spectrum disorder [ASD]). Although a small number of children may experience less stress or anxiety due to reduced social and academic expectations,¹ for many children with NDDs, and particularly those with ASD, carefully developed behavioral and environmental supports, and consistent and predictable routines and expectations, are vital for their mental well-being.² Consequently, abrupt discontinuation of these supports during quarantine and prolonged isolation creates a real risk for behavioral exacerbations in this vulnerable population.³⁻⁶ Possible consequences for family members include physical and mental strain,⁷ whereas for the child with an NDD, increased emotional distress and challenging behavior may create safety concerns and the need for hospitalization.^{4,6} Children with NDDs may be at increased risk for COVID and COVID-related complications,⁸ emphasizing the need for preventive and/or crisis behavioral health care availability outside of emergency and hospital settings.

TABLE 1 Key Components of the COVID-19 Neurodevelopmental Disorders (NDD) Brief Intervention Program

Component	Description
Target population	Families of children with a neurodevelopmental disorder (eg, autism spectrum disorder, intellectual disability, attention-deficit/hyperactivity disorder [ADHD], learning disorder) experiencing an acute exacerbation in behavioral/mental health difficulties in the context of the COVID-19 pandemic, residing in the province of Ontario
Multidisciplinary assessment	A child psychiatrist, a psychologist, and a behavioral expert (behavior therapist or social worker with experience in NDD behavioral intervention)
Optimal use of technology	Virtual consultation, with families and clinicians all joining remotely via an online platform. Parallel text messaging thread using WhatsApp by clinical team to facilitate 1 lead speaker. Shared online document for simultaneous drafting of report during assessment
Focused assessment of problem, context, and safety concerns	Discuss current behavioral/mental health issues; consider caregiver exhaustion and possible safety concerns. Review child's background, medical and developmental history, context, home environment and structure, usual and current supports.
Immediate feedback and rapid turn-around report addressed to the family	After assessment, families briefly sign off, allowing the team to debrief and to develop a plan. Subsequently, families sign on to receive feedback. A short report is written addressed to the families (copy to referring physician), formulated in lay language and with sufficiently detailed recommendations. Turn-around within 24 to 48 hours via e-mail communication (provided parental permission)
Common recommendations/interventions	Suggestions to adapt, simplify or develop daily activity schedule Safety assessment and crisis plan Development of a focused behavioral treatment plan, with consistent rewards and/or responses to specific behaviors Development of beginner communication systems Parent support, validation, connection to resources Short-term medication adjustment/start Brief follow-up in selected cases

During the COVID-19 pandemic, many parents of children with an NDD have identified a need to access specialized advice to help them cope with their child's behavioral and mental health challenges and to prevent existing problems from worsening.⁹ Working within the restrictions imposed by the pandemic creates a challenge for health care providers, but also provides an opportunity to use virtual technology to address the immediate needs of these families.¹⁰ With this in mind, we developed a novel, virtual, multidisciplinary, brief intervention program that was rapidly deployed at the Hospital for Sick Children (SickKids) in Toronto, Ontario, Canada. The "COVID-19 NDD clinic" was developed in the context of temporary closures and a shift to home-based virtual consultation because of the pandemic. Early in the pandemic, provinces across Canada introduced new consultation billing rules outlining financial coverage for online health care consultations. Hence, in Ontario, the service is available without charge to families of children under the Ontario Health Insurance Plan. It provides timely and practical assistance to families who are seeking help regarding how to manage their child's behavior during the pandemic when other services and supports are not available.

By intervening quickly, we hope to provide an alternative to families who may otherwise be forced to seek help for behavioral or mental health concerns through hospital emergency departments or inpatient admissions. In addition to its online operation, the COVID-19 NDD Clinic has several distinct features: (1) each assessment is performed by a multidisciplinary team; (2) the approach is strictly solution based, focusing on the immediate problem at hand; and (3) the report, including recommendations, is written in lay language and addressed primarily to parents. These features have proved both efficient and highly appreciated both by families and by referring physicians. Here, we provide a short description of this clinic and detailed operational procedures (Supplement 1, available online) as well as results of an evaluation survey. We anticipate that our strategy may be adoptable by other centers to address some of the most pressing needs of families of children with NDDs.

AIMS AND METHODS OF THE COVID-19 NDD CLINIC

There are 2 main aims for the assessment. The first aim is to identify the families' most pressing concern(s) regarding their

TABLE 2 Caregiver/Family (n = 20), and Referring Physician (n = 19) Feedback on the COVID-19 Neurodevelopmental Disorders (NDD) Brief Intervention Program

Caregiver and family quality improvement survey items	Agree or strongly agree
I was satisfied with the wait time from referral to appointment.	20
Main concerns regarding my child's behavior ... were identified.	20
Recommendations were practical	20
I feel better prepared to handle my child's behavior	12
I was able to put recommendations into practice	15
Recommendations were helpful	17
Written report was helpful	19
Online experience was easy	19
Online tutorial with clinic administrator before the visit was helpful	20
Referring physician quality improvement survey items	Agree or strongly agree
Overall assessment/recommendations were helpful	19
In my practice, the level of need for a service such as the COVID-19 Virtual Clinic is high during this pandemic	19
I feel that a similar problem-focused virtual consultation for behavioral issues of children with developmental disorders is needed beyond this pandemic	19
Addressing the report directly to families is much better	15

child's behavior during COVID-19, and to determine whether there are potential safety issues that must be addressed. In addition to the child's status, information is gathered regarding the mental well-being of caregivers and siblings with special attention to the possibility of exhaustion and safety concerns. The second aim is to gather background and contextual information that may have a bearing on the current situation, including current and past medical and mental health conditions and treatments, previous and current supports, and, importantly, current environmental factors.

A noteworthy aspect of the methodological approach is that the clinic is multidisciplinary: each assessment is carried out by a child and adolescent psychiatrist, psychologist, and behavioral expert (Table 1). This approach allows for a highly efficient "ad hoc" integrated evaluation of medical, psychiatric, behavioral, and psychological aspects of the presented problems as well as possible recommendations that take these different perspectives fully into account.

Furthermore, the virtual assessments focus on rapid clarification of the problematic behavior in a relatively brief time span (approximately 60–90 minutes). During this time, context is explored as well, including strengths and challenges of the child and family. This focused approach implies that the consultation forgoes an in-depth formal diagnostic assessment, as this would typically require a more comprehensive evaluation while not necessarily addressing the immediate needs of the families in the current context.

After completion of the assessment, the ultimate objective is to provide a formulation that integrates family and clinical perspectives, and to offer recommendations about how to manage the current situation. If necessary, safety concerns are addressed explicitly. Recommendations are addressed to parents and formulated in lay language. Care is taken that they are specific and realistically attainable, and tailored to the family's strengths, challenges, and context. Suggestions are offered about how to prevent or to minimize the behavioral concerns, in addition to sticking to a plan and having realistic expectations within the limitations imposed due to COVID-19 and the family situation. Medication recommendations are discussed with the family and provided to the treating physician, keeping in mind that in the current situation, the necessity of medication and the potential risk of side effects (such as oversedation) may be weighed differently than in the regular, pre-COVID-19 context. The importance of parents attending to their own emotional and physical needs is emphasized in order to prevent exhaustion and burnout. The COVID-19 NDD clinic team does not replace previous or ongoing care provided by other professionals, but can liaise with existing care providers who can support the implementation of recommendations.

EVALUATION OF COVID-19 NDD CLINIC

Between March 31 and June 8, 2020, a total of 43 families received consultation through the COVID-19 NDD Clinic

(representing 4–5 referrals per week); appointments were scheduled within 5 days of receiving the referral. A total of 24 referrals were received from clinicians external to The Hospital for Sick Children. The majority of referrals ($n = 36$) were for COVID-19–related issues (eg, increase in child's aggression and outbursts, regression in functioning during lockdown); in 7 cases it became clear during the assessment that parents were actually seeking a diagnostic or psychoeducational assessment, not specifically related to the pandemic. Referred children ranged in age from 3 to 18 years and with the exception of 2 children; all had a previous diagnosis of an NDD. Safety concerns were identified for 20 children (eg, aggression to self or others causing injury or risk of elopement), and 1 family was assisted to access emergency services following the assessment. Medication recommendations were provided to treating physicians for 24 children seen, and community-based behavioral services were recommended to 10 families. The vast majority did not require follow up by the assessment team; only for 2 families did the team offer check-in calls for a limited (2-week) period of time after the assessment.

Evaluation of the clinic occurred via an anonymous survey sent to families and referring physicians. This quality improvement initiative was approved by the Hospital for Sick Children. In total, 20 families and 19 clinicians responded (Table 2). Survey responses and open-ended feedback indicated that the program was well received, the virtual format and technology ran smoothly, and the recommendations were generally perceived as helpful. Two of the suggestions offered in feedback were a request for follow-up appointments to support implementation of strategies and referrals for parents to receive counseling and support. Referring clinicians identified a need for continuity of this type of initiative beyond pandemic times.

LESSONS LEARNED AND NEXT STEPS

There were several challenges related to the implementation of this program, including ensuring that referrals were motivated by COVID-19–related problems (for 7 of the 43 referrals, the assessment revealed that this was not the case), as well as the challenge of covering sometimes-complex issues in a relatively short period of time. Nevertheless, we have found overall the COVID-19 NDD brief intervention program to be a feasible and efficient way to respond to the needs of a vulnerable population differentially affected by restrictions during the pandemic. In addition to supporting the NDD community and care providers, participating clinicians report benefits from working together, including expanding their range of recommendations, building confidence in related disciplines, and feeling supported in managing complex and potentially high-risk clinical situations virtually.

Often, the most frequent therapeutic interventions included parent support/validation given the exceptional circumstances, as well as expectation management in the setting of chronicity and complexity. Overall, this type of clinic may be helpful for intermittent deployment/expansion as the stages of restriction (in particular, school closures) are lifted or re-imposed while the pandemic evolves. In addition, this type of initiative may prove beneficial for clinical institutions/programs beyond the pandemic setting, to support families and to manage wait-lists and acuity in the setting of scarce resources, until a more comprehensive assessment or definitive treatment plan can be arranged.

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School Closures and Social Anxiety During the COVID-19 Pandemic



To the Editor:

In their recent systematic review, Loades *et al.*¹ reported on the effects that social isolation and loneliness may have on children and adolescents during the global 2019 novel coronavirus disease (COVID-19) pandemic, with their findings suggesting associations between social anxiety and loneliness/social isolation. While this is undoubtedly true for many children and adolescents, it is also worth commenting on the subset of children and youths with social phobia for whom a temporary lessening of distress may be observed while schools are closed owing to a lack of exposure to anxiety-provoking situations in the school environment.

Exposure to feared situations is generally regarded as an essential component of anxiety treatment.² School environments typically provide a plethora of opportunities for social exposure. Needless to say, opportunities for exposure to social situations have been exceedingly limited for many children and youths during the COVID-19 outbreak. Furthermore, in many jurisdictions, citizens are being commended by politicians and public health officials for engaging in practices aimed at mitigating viral spread, including physical distancing and avoiding nonessential travel outside of their homes.³ As a result, people with social anxiety disorder are receiving

positive reinforcement for avoiding the very situations to which psychological treatment would encourage exposure. In addition, avoidance begets further avoidance by way of negative reinforcement,⁴ and acute destabilization for socially anxious children and youths is foreseeable when schools reopen and attendance is expected. Loades *et al.*¹ aptly acknowledged the limitations of extrapolating existing evidence to the current context. Indeed, the mixture of limited opportunities for social exposure combined with aspects of both positive and negative reinforcement for avoidance of exposure highlights the uniqueness of circumstances resulting from the COVID-19 pandemic.

Despite pragmatic challenges imposed by the COVID-19 outbreak, it is imperative that treatment of social anxiety disorder continue during the pandemic. Clinicians employing cognitive-behavioral therapy may consider shifting their interventions to address negative beliefs and processing of events, rather than emphasizing exposure. Creative and innovative treatment strategies are also called for. Loades *et al.*¹ suggested that digital interventions may facilitate treatment for children and adolescents during times of social isolation. Indeed, health care providers in various fields are increasingly turning to virtual care technologies to limit physical meetings between patients and clinicians to mitigate viral spread.⁵ In treating social anxiety, clinicians should consider exposure to social and performance situations using virtual means.

Clinicians may be lulled into a false sense of security when socially anxious children and adolescents report improvement concurrent with school closures. However, this does not obviate the need for continued treatment during the pandemic. Such improvement is likely to be short-lived, and children and youths with social anxiety disorder will face significant challenges when schools reopen, highlighting the need for continued treatment during the pandemic.

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